

Residual Scalar Norm Attainment from an Isolated Norming Orbit

FA/Banach discovery run `fa_banach_001`

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Status

This is a strengthened partial packet for arXiv:2203.04023. It incorporates the supplied report `evidence/supplied_residuality_isolated_norming_2026_06_22/`.

The report repairs the previous packet's nonseparable category step and extends the finite-boundary result to ranges with one isolated norming orbit. The full question of Jung–Martin–Rueda Zoca remains open here.

Source and Question

The source paper is:

Mingu Jung, Miguel Martin, Abraham Rueda Zoca, *Residuality in the set of norm attaining operators between Banach spaces*, arXiv:2203.04023.

For Banach spaces X and Y over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, write

$$\text{NA}(X, Y) = \{T \in \mathcal{L}(X, Y) : \text{there is } x \in S_X \text{ with } \|Tx\| = \|T\|\}.$$

Remark 4.2 asks whether residuality of $\text{NA}(X, Y)$ in $\mathcal{L}(X, Y)$, for some nonzero range Y , forces residuality of scalar $\text{NA}(X, \mathbb{K})$ in X^* .

where we have used (1) and (2) in the last inequality. This contradicts (3). So, we conclude that $A_{m,k}$ is nowhere dense for each $m, k \in \mathbb{N}$. $\square$

Remark 4.2. It is not possible, in general, to get denseness of Fréchet differentiable points of $\mathcal{L}(X, Y)$ when $\text{NA}(X, Y)$ is residual. For instance, if X has the Radon-Nikodým property, then $\text{ASE}(X, \ell_1)$ is dense in $\mathcal{L}(X, \ell_1)$ (hence, $\text{NA}(X, \ell_1)$ is residual), but there is no point in $\mathcal{L}(X, \ell_1)$ where the norm on $\mathcal{L}(X, \ell_1)$ is Fréchet differentiable since the norm of ℓ_1 is nowhere Fréchet differentiable (use Proposition 1.6). On the other hand, there is no known objection, as far as we know, to get the denseness of $\text{ASE}(X, Y)$ from the residuality of $\text{NA}(X, Y)$ in complete generality. On the other hand, we do not know if the residuality of $\text{NA}(X, Y)$ for a non-trivial Y implies that of $\text{NA}(X, \mathbb{K})$ as the denseness of $\text{ASE}(X, Y)$ does.

Remark 4.3. (1) If the Banach space Y in Theorem 4.1 is *reflexive*, then the denseness of the set of Fréchet differentiable points of $\mathcal{L}(X, Y)$ can be obtained directly from [47, Theorem 3.1]. Indeed, in this case we have that $(X \widehat{\otimes}_{\pi} Y^*)^* = \mathcal{L}(X, Y)$ and the inclusion $\text{NA}(X, Y) \subseteq \text{NA}(X \widehat{\otimes}_{\pi} Y^*, \mathbb{K})$ holds. It follows that $\text{NA}(X \widehat{\otimes}_{\pi} Y^*, \mathbb{K})$ is residual.

The source paper gives positive results under additional hypotheses. In particular, if X and Y^* are separable, residuality of $\text{NA}(X, Y)$ implies density of $\text{ASE}(X, Y)$ [4, Theorem 4.1]; density of $\text{ASE}(X, Y)$ implies density of the strongly exposing functionals on X [4, Proposition 1.5]. The unrestricted implication in Remark 4.2 is not settled by the source paper.

Classification

This packet is a candidate partial result, likely valid pending human review. It proves the scalar-residuality conclusion for a substantial class of ranges with an isolated norming orbit. It is not a full solution or a counterexample to Remark 4.2.

The previous packet proved the finite-boundary and real polyhedral subcases. The supplied report found one real gap in that writeup: the usual Kuratowski–Ulam section theorem was invoked without a separability or countable- π -base hypothesis. The argument only needs the much more special cylinder-reflection theorem below, which is valid for completely metrizable spaces. The active proof below uses that repair and explicitly records the equality $\|T\| = \|\phi_0 \circ T\|$ on the domination region.

Cylinder Reflection

Lemma 1 (Cylinder reflection). *Let P and Q be completely metrizable spaces, with $P \neq \emptyset$. For every $A \subset Q$,*

$$A \text{ is meagre in } Q \iff P \times A \text{ is meagre in } P \times Q.$$

Consequently, if U and V are nonempty open subsets of Banach spaces and $A \subset U$, then

$$A \times V \text{ residual in } U \times V \implies A \text{ residual in } U.$$

Proof. The equivalence is the cylinder theorem for completely metrizable spaces; see [2, Lemma 4.3]. Open subsets of Banach spaces are completely metrizable in compatible complete metrics. Applying the equivalence to the complement of A gives the final assertion. $\square$

Proof Intuition

The key point is to work on open cylinders, not on a rank-one slice. If (ϕ_0, y_0) isolates one norming orbit of Y , then every operator in a suitable open region can be written as $T = y_0 \otimes f + H$, and the ϕ_0 -coordinate strictly dominates all other norming functionals. In that region T attains its norm exactly when the scalar functional $\phi_0 \circ T = f$ attains its norm. Residuality in the operator space therefore reflects to residuality on scalar annuli by the cylinder lemma.

Main Partial Theorem

Definition 1. *A norming set $\Gamma \subset S_{Y^*}$ has an isolated norming orbit if there are $\phi_0 \in \Gamma$ and $y_0 \in S_Y$ such that*

$$\phi_0(y_0) = 1, \quad \beta := \sup\{|\gamma(y_0)| : \gamma \in \Gamma \setminus \mathbb{T}\phi_0\} < 1, \quad (1)$$

where $\mathbb{T} = \{\lambda \in \mathbb{K} : |\lambda| = 1\}$ and $\sup \emptyset = 0$. The set Γ is norming when

$$\|y\| = \sup_{\gamma \in \Gamma} |\gamma(y)| \quad (y \in Y).$$

Theorem 1. *Let X and $Y \neq \{0\}$ be Banach spaces, and suppose that Y has a norming set Γ with an isolated norming orbit witnessed by (ϕ_0, y_0) . Let $\mathcal{I} \subset \mathcal{L}(X, Y)$ be a closed linear subspace such that*

$$y_0 \otimes f \in \mathcal{I} \quad (f \in X^*). \quad (2)$$

If $\text{NA}(X, Y) \cap \mathcal{I}$ is residual in $\mathcal{I}$, then $\text{NA}(X, \mathbb{K})$ is residual in X^ .*

Proof. Set

$$\mathcal{I}_0 = \{H \in \mathcal{I} : \phi_0 \circ H = 0\}.$$

The map

$$\Psi : X^* \times \mathcal{I}_0 \longrightarrow \mathcal{I}, \quad \Psi(f, H) = y_0 \otimes f + H \quad (3)$$

is a linear homeomorphism. Its inverse is

$$T \longmapsto (\phi_0 \circ T, T - y_0 \otimes (\phi_0 \circ T)).$$

Every $H \in \mathcal{I}_0$ takes values in $Z = \ker \phi_0$.

Put

$$C = \sup\{\|\gamma|_Z\| : \gamma \in \Gamma \setminus \mathbb{T}\phi_0\} \leq 1,$$

again taking $C = 0$ if the indexing set is empty. Define the open domination region

$$\mathcal{W} = \{\Psi(f, H) : (1 - \beta)\|f\| > C\|H\|\}. \quad (4)$$

Fix $T = \Psi(f, H) \in \mathcal{W}$. If $\gamma \in \Gamma \cap \mathbb{T}\phi_0$, then $\|\gamma \circ T\| = \|f\|$. If $\gamma \notin \mathbb{T}\phi_0$, then

$$\|\gamma \circ T\| \leq |\gamma(y_0)|\|f\| + \|\gamma|_Z\|\|H\| \leq \beta\|f\| + C\|H\| < \|f\|. \quad (5)$$

Since Γ norms Y ,

$$\|T\| = \sup_{\gamma \in \Gamma} \|\gamma \circ T\| = \|f\|. \quad (6)$$

We claim that on $\mathcal{W}$,

$$T \in \text{NA}(X, Y) \iff f \in \text{NA}(X, \mathbb{K}). \quad (7)$$

If f attains its norm at $x \in S_X$, then

$$\|Tx\| \geq |\phi_0(Tx)| = |f(x)| = \|f\| = \|T\|,$$

so T attains its norm.

Conversely, suppose that $\|Tx\| = \|T\| = \|f\|$ for some $x \in S_X$. By (5),

$$\sup_{\gamma \in \Gamma \setminus \mathbb{T}\phi_0} |\gamma(Tx)| < \|f\|.$$

For $\gamma \in \Gamma \cap \mathbb{T}\phi_0$, one has $|\gamma(Tx)| = |f(x)|$. Because

$$\|Tx\| = \sup_{\gamma \in \Gamma} |\gamma(Tx)| = \|f\|,$$

it follows that $|f(x)| = \|f\|$. This proves (7).

Let $A = \text{NA}(X, \mathbb{K})$, and for $m \geq 1$ let

$$U_m = \{f \in X^* : \|f\| > 1/m\}.$$

If $C > 0$, set $r_m = (1 - \beta)/(2Cm)$; if $C = 0$, choose any $r_m > 0$. Then

$$U_m \times B_{\mathcal{I}_0}(0, r_m) \subset \Psi^{-1}(\mathcal{W}).$$

Residuality of $\text{NA}(X, Y) \cap \mathcal{I}$, restriction to this open cylinder, and (7) show that

$$(A \cap U_m) \times B_{\mathcal{I}_0}(0, r_m)$$

is residual in $U_m \times B_{\mathcal{I}_0}(0, r_m)$. By Lemma 1, $A \cap U_m$ is residual in U_m . Hence $U_m \setminus A$ is meagre in X^* for every m . Since

$$X^* \setminus \{0\} = \bigcup_{m=1}^{\infty} U_m$$

and $0 \in A$, the complement of A is meagre in X^* . Thus $\text{NA}(X, \mathbb{K})$ is residual in X^* . $\square$

Consequences

Corollary 1 (Finite-boundary ranges). *Suppose that a finite set $\Phi \subset S_{Y^*}$ satisfies*

$$\|y\| = \max_{\phi \in \Phi} |\phi(y)| \quad (y \in Y),$$

and that some $\phi_0 \in \Phi$ admits $y_0 \in S_Y$ with

$$\phi_0(y_0) = 1, \quad \max_{\phi \in \Phi \setminus \mathbb{T}\phi_0} |\phi(y_0)| < 1.$$

If $\text{NA}(X, Y)$ is residual in $\mathcal{L}(X, Y)$, then $\text{NA}(X, \mathbb{K})$ is residual in X^ .*

Proof. Apply Theorem 1 with $\Gamma = \Phi$ and $\mathcal{I} = \mathcal{L}(X, Y)$. □

Corollary 2 (Real polyhedral ranges). *Let Y be a nonzero finite-dimensional real polyhedral Banach space. If $\text{NA}(X, Y)$ is residual in $\mathcal{L}(X, Y)$, then $\text{NA}(X, \mathbb{R})$ is residual in X^* .*

Proof. Choose one representative from each opposite pair of vertices of B_{Y^*} . These representatives form a finite norming set. Every vertex of a finite-dimensional polytope is exposed. If ϕ_0 is exposed by y_0 , normalize so that $\phi_0(y_0) = 1$. Exposure of both ϕ and $-\phi$ gives $|\phi(y_0)| < 1$ for every other orbit. Corollary 1 applies. □

Corollary 3 (Property quasi- β). *Let $Y \neq \{0\}$ have property quasi- β . If $\text{NA}(X, Y)$ is residual in $\mathcal{L}(X, Y)$, then $\text{NA}(X, \mathbb{K})$ is residual in X^* .*

Proof. Use the quasi- β data $(\mathcal{A}, \sigma, \rho)$. The set $\mathcal{A} \subset S_{Y^*}$ is norming by the weak-star closure condition on the extreme points of B_{Y^*} , as in [4, Section 3.1]. Fix $a_0 \in \mathcal{A}$ and put $y_0 = \sigma(a_0)$. No distinct rotation of a_0 can lie in $\mathcal{A}$, since it would have modulus one at y_0 , contradicting the quasi- β inequality. Hence

$$\sup_{a \in \mathcal{A} \setminus \mathbb{T}a_0} |a(y_0)| \leq \rho(a_0) < 1.$$

Thus $\mathcal{A}$ has an isolated norming orbit, and Theorem 1 applies. □

Theorem 2 (Countably many extreme orbits). *Let $Y \neq \{0\}$ be finite-dimensional and suppose that*

$$\text{ext}(B_{Y^*})/\mathbb{T}$$

is countable. Then Y has a norming set with an isolated norming orbit. Consequently, for every Banach space X ,

$$\text{NA}(X, Y) \text{ residual in } \mathcal{L}(X, Y) \implies \text{NA}(X, \mathbb{K}) \text{ residual in } X^*.$$

Proof. Put $K = B_{Y^*}$ and $E = \text{ext}(K)$, regarding Y^* as a real finite-dimensional normed space for convexity. The set E is a G_δ subset of K : $K \setminus E = \bigcup_n F_n$, where F_n consists of points that are midpoints of two points of K at distance at least $1/n$. Each F_n is compact. Thus E is completely metrizable and is a Baire space.

By hypothesis, E is the union of countably many compact $\mathbb{T}$ -orbits. If every orbit had empty interior in E , E would be meagre in itself. Thus one orbit $\mathcal{O} = \mathbb{T}\phi_0$ is open in E .

Let $F = \overline{E \setminus \mathcal{O}}^K$. Then $F \cap \mathcal{O} = \emptyset$. If $E \setminus \mathcal{O} = \emptyset$, any norm-one vector where ϕ_0 attains its norm gives the desired isolated orbit. Otherwise set $C = \text{co}(F)$. The compact convex set C is $\mathbb{T}$ -invariant and does not contain ϕ_0 ; otherwise the extremality of ϕ_0 would force a point of F to lie in $\mathcal{O}$.

Strict separation gives $y \in Y$ such that

$$\operatorname{Re} \phi_0(y) > \sup_{\psi \in C} \operatorname{Re} \psi(y).$$

In the complex case, finite dimensionality identifies real-linear functionals on Y^* with maps $\psi \mapsto \operatorname{Re} \psi(y)$. Since C is $\mathbb{T}$ -invariant,

$$s := \sup_{\psi \in C} \operatorname{Re} \psi(y) = \sup_{\psi \in C} |\psi(y)|.$$

Rotate ϕ_0 so that $a = \phi_0(y) = |\phi_0(y)| > s$. Since E is norming and every extreme point belongs either to $\mathcal{O}$ or to $F \subset C$, $\|y\| = a$. Set $y_0 = y/a$. Then

$$\phi_0(y_0) = 1, \quad \sup_{\psi \in E \setminus \mathcal{O}} |\psi(y_0)| \leq s/a < 1.$$

Choosing one representative from each $\mathbb{T}$ -orbit in E , including ϕ_0 , gives a norming set with an isolated norming orbit. The final implication is Theorem 1. $\square$

Combining Theorem 2 with [4, Example 3.18(a)] gives, for finite-dimensional ranges with countably many extreme dual orbits, an equivalence between residuality of $\operatorname{NA}(X, Y)$ in $\mathcal{L}(X, Y)$ and residuality of $\operatorname{NA}(X, \mathbb{K})$ in X^* .

Why the Full Question Remains Open

The proof uses a genuine geometric gap. If Y is smooth and its norming functionals vary continuously, there need not be an isolated norming orbit. The real Euclidean plane is the basic example: its dual extreme points form a circle, and no orbit modulo signs is isolated. In that setting a small transverse perturbation can rotate the norming functional, so the fixed coordinate domination used above is unavailable.

Several natural routes stop here. A residual subset of $\mathcal{L}(X, Y)$ cannot be safely restricted to the closed rank-one slice $\{y_0 \otimes f : f \in X^*\}$. Approximation by nearby norm-attaining scalar coordinates gives only Bishop–Phelps density, not residuality. Compactness may produce norming functionals $\phi_n \rightarrow \phi$, but the corresponding scalar norm-attainers $T^* \phi_n$ can converge to a non-norm-attaining functional. The uniform gap in (1) is exactly what prevents that loss in the theorem above.

Evidence, Novelty, and Review Recommendation

The supplied report states that searches through June 21, 2026 by exact wording, title citations of [4], and recent work on norm-attaining and range-strongly-exposing operators did not locate a proof or counterexample to the full question. This packet therefore records a research-level partial candidate, not a peer-reviewed theorem and not a claim of full novelty.

Human review should focus on:

- the cylinder-reflection theorem as used in Lemma 1;
- the equality $\|T\| = \|\phi_0 \circ T\|$ on $\mathcal{W}$;
- the property quasi- β norming-set argument;
- the Baire argument selecting an open extreme-point orbit in Theorem 2.

No computational verification is needed.

References

- [1] M. D. Acosta, F. J. Aguirre and R. Paya, *A new sufficient condition for the denseness of norm attaining operators*, Rocky Mountain J. Math. **26** (1996), 407–418.
- [2] W. Brian, *Cardinal invariants of a meager ideal*, Topology Appl. **379** (2026), Article 109516.
- [3] G. Choi, H. del Rio, A. Fovelle, M. Jung and M. Martin, *Range strongly exposing operators between Banach spaces*, arXiv:2503.18581v2 (2025).
- [4] M. Jung, M. Martin and A. Rueda Zoca, *Residuality in the set of norm attaining operators between Banach spaces*, J. Funct. Anal. **284** (2023), Article 109746.
- [5] V. Kadets, G. Lopez, M. Martin and D. Werner, *Norm attaining operators of finite rank*, in *The Mathematical Legacy of Victor Lomonosov*, Advances in Analysis and Geometry, vol. 2, de Gruyter, 2020, 157–187.
- [6] M. Martin, *A Banach space whose set of norm-attaining functionals is algebraically trivial*, J. Funct. Anal. **288** (2025), Article 110815.